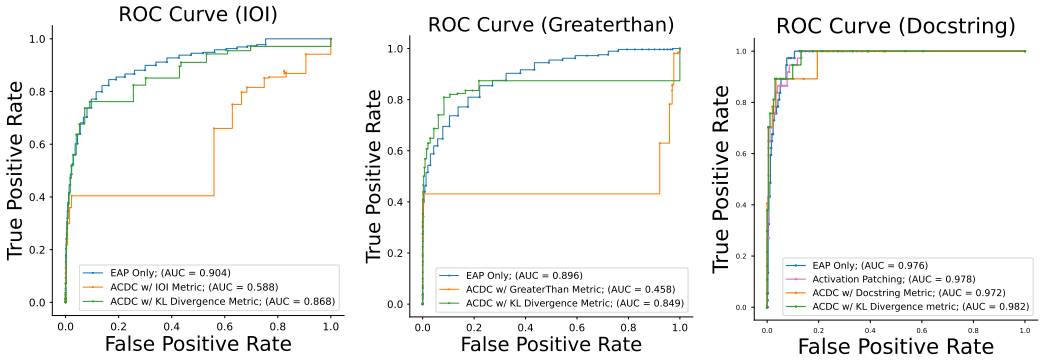
4 Results

4.1 Edge Attribution Patching vs Activation Patching vs ACDC

We compare Edge Attribution Patching (EAP) and ACDC on the Indirect Object Identification (IOI), Docstring, and Greater-Than tasks. For each of these tasks, previous studies identified a subgraph (circuit) relevant for solving the task. We use their results as a ground truth for benchmarking both methods. We also compare using ACDC with the task-specific metrics (e.g logit difference) and KL Divergence (which was originally recommended). For the docstring task, we also include repeated activation patching as another point of reference for performance comparisons. We applied repeated activation patching by running the same circuit discovery method described in Section 3.3 but using Equation (1) rather than absolute attribution scores. Activation patching was not included in the other tasks as it was too computationally expensive to run on the GPT-2 small models used by IOI and Greater-Than. Subnetworks found using EAP for all three tasks are shown in Appendix A.



(a) IOI task (b) Greater-Than task (c) Docstring task

Figure 2: ROC Curves comparing EAP, ACDC with task metric, and ACDC with KL Divergence. The Docstring plot also compares to Activation Patching.

The ROC curves in Figure 2 suggest the performance of EAP is better than ACDC overall: it has the maximal AUC in Figure 2a-2b, while ACDC used with the KL Divergence metric outperforms EAP in Figure 2c. ACDC outperformed the existing methods HISP and Subnetwork Probing methods [10]. We conclude EAP outperforms all previous methods for circuit discovery, since it is competitive with ACDC on recovering circuits while significantly reducing the computational demand: EAP only takes a constant number of forward and backwards passes while the number of forward passes required by ACDC is scaling exponentially with the number of nodes.

4.2 Validating EAP Attribution Scores

In this section, we look at the approximate metric change (attribution score) EAP assigns to each edge in the model. We aim to understand the relation between the attribution score and the function of the edge in the task being studied. First, we look at the distribution of scores for edges in the circuit compared to edges not in the circuit for each of the three tasks.

Figure 3 shows the distribution of attribution scores for the IOI task. The distributions for the remaining tasks can be found in Appendix B. Qualitatively, attribution scores for edges in the circuit tend to be spread further from zero. Furthermore, there are only 6 edges outside of the interval $[-0.25, 0.25]$ that aren't part of the IOI circuit. We further explore the attribution scores for the IOI circuit's classes of heads in Appendix E.

5 Limitations

We introduced edge activation patching as an approximation to activation patching. However, we found that edge activation patching outperformed ACDC, a technique based on activation patching (Section 4). In this section, we investigate whether attribution patching's success is due to extremely

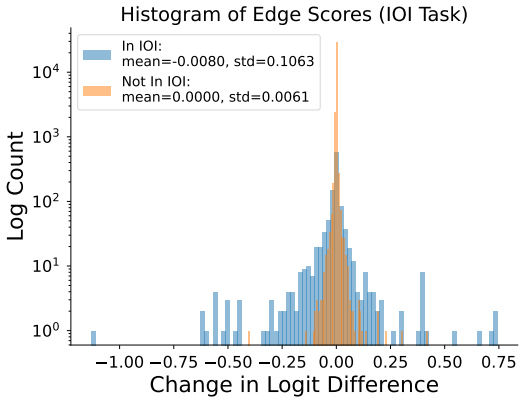


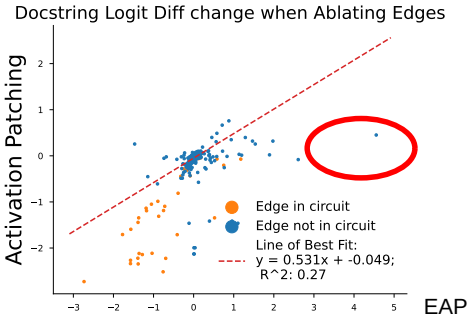
Figure 3: Distribution of Attribution Scores for the IOI Task (Logit Diff)

accurate approximations (in Section 5.1 we find that the answer is no), and whether there is any further use for ACDC (in Section 5.2 we find that the answer is yes). We use the docstring task as a case study due to the small model size used.

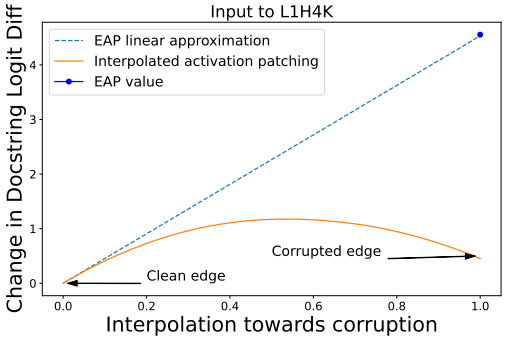
5.1 How faithful are Attribution Patching's approximations?

To study how faithful the approximation Equation (2) is, we plot the attribution patching scores (Equation (2)) against the activation patching scores (Equation (1)) in Figure 4a. Surprisingly, we find a fairly weak correlation between activation and attribution patching scores ($R^2 = 0.27$). Further, the line of best fit has gradient 0.531, suggesting that attribution patching estimates the effect of activation patching as twice as important as it really is.

Moreover, we can gain some sense for the discrepancy between activation and attribution patching by studying the continuous transition between clean (e_{clean}) and corrupted (e_{corr}) activations in Equation (1), i.e studying the values $|L(x_{\text{clean}}|\ \text{do}(E = \lambda e_{\text{corr}} + (1-\lambda)e_{\text{clean}})) - L(x_{\text{clean}})|$ for $0 \leq \lambda \leq 1$. We can compare this to the linear approximations of Attribution Patching $\lambda \Delta_e L$. Figure 4b shows the result for one edge in the docstring circuit where the linear approximation to activation patching is not accurate.



(a) Distribution of attribution scores for edges from activation patching and attribution patching. Circled: outlier EAP point studied in Figure 4b.



(b) Visualizing the rightmost point in Figure 4a. Note that corrupting this edge (surprisingly) slightly increases the logit difference on the Docstring task (higher logit difference is better). However, EAP overestimates how large this increase is.

Figure 4: Visualizing Edge Attribution Patching.

We find that interpolating towards the corrupted input creates a concave curve (Figure 4b) such that the linear approximation at $\lambda = 0$ overestimates the effect of activation patching this edge. In Appendix D we show that this also holds for the other outlier edges in the ellipse in Figure 4a.

5.2 Is there any further use for ACDC?

In Section 5.1 above, we found that EAP overestimates activation patching in cases where the attribution score is concave. This suggests the potential to refine the result by running ACDC on the pruned subgraph returned by EAP. We ran EAP first, then ACDC on the resulting subgraph for the Docstring task, varying pruning thresholds for EAP and ACDC independently. Figure 5 compares the TPR and FPR for the combined methods with the ROC curve of EAP only. The combined methods show increased performance compared to EAP only.

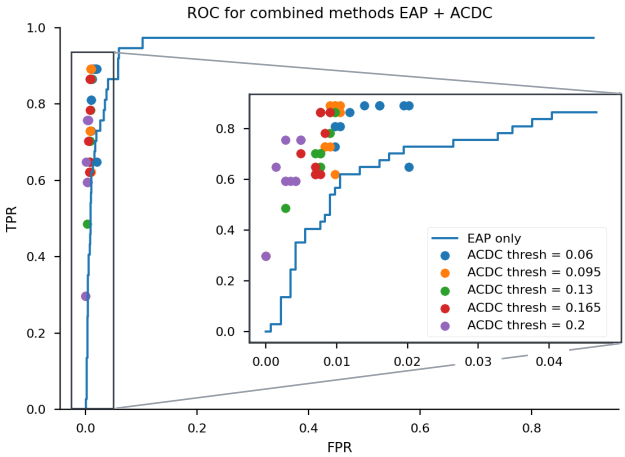


Figure 5: Comparing statistics of the combined EAP + ACDC methods with EAP only. The inset shows a zoom to the significant area of the statistics of the combined method.

Finally, one further limitation of this research is that the metrics used for interpretability do not precisely capture meaningful human understanding. Recovering a subgraph that humans previously recovered is limited because i) we can't evaluate this metric for interpretability tasks that we don't yet understand and ii) human-found circuits are imperfect, increasing the noise in this measurement.

6 Conclusion

We provide evidence that Edge Attribution Patching (EAP) outperforms ACDC in identifying circuits while being substantially faster to compute. This result is surprising, as EAP is an approximation for activation patching, the method applied by ACDC. However, running ACDC on the prepruned subnetwork found by EAP can improve the identification of relevant edges. Therefore, we suggest future circuit discovery experiments to run EAP first and then apply ACDC.

7 Author Contributions

Aaquib Syed and Can Rager proposed combining ACDC with attribution patching methods and implemented initial prototypes. Arthur Conmy advised working on attributing edges rather than nodes and Aaquib made the first findings that this outperformed Automatic Circuit Discovery. All authors worked on the paper's figures, experiments and code.

8 Acknowledgements

We would like to thank Callum McDougall for organising ARENA and providing a great introduction to interpretability, and Rusheb Shah and Lucy Farnik for collaboration on the ARENA hackathon prototype which this work is based on. We would like to thank Neel Nanda for a helpful discussion and János Kramár, Stephen Casper and Euan Ong for suggestions based on a earlier version of this work.